Channel Flow of a Compressible Fluid & Stagnation Enthalpy

AA 504 (Compressible Fluid Mechanics) Autumn 2026, UW

Assignment: Homework #1

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```
clear % Clears workspace
clc % Clears command window
close all % Closes all figures
```

Problem 1. Channel Flow of a Compressible Fluid

Problem Statement

Based off of the movie "Channel Flow of a Compressible Fluid" [1], write one paragraph on the most counter-intuitive aspect of the physics to you.

```
imshow('HW1_Prob1.png');
```



Statement

The most counter-intuitive aspect of the physics in channel flow of a compressible fluid is the *area-velocity relation*. The rules "flip" when moving from subsonic speeds (Mach < 1) to supersonic speeds (Mach > 1). We live in the subsonic world where a decrease in area gives an increase in velocity, such as when pinching a garden hose (area decrease) makes the water shoot out faster. But once the flow goes supersonic, the rules are completely reversed; velocity is only increased by increasing the area.

These Mach number effects are mathematically described by the area-velocity relation [3]:

$$\frac{du}{u} = -\frac{dA/A}{1-M^2} \tag{2.27}$$

This is visualized by plotting equation (2.27):

```
%% Area changes and nozzle speed -- Liepmann & Roshko, Eq. 2.27
% du/u = -(dA/A)/(1-M^2)
% Assumptions: steady, quasi-1D, isentropic, calorically perfect gas;
% no shocks, heat addition, friction, or shaft work.
% We solve the integrated area-Mach relation rather than divide by
% (1-M^2) at the sonic throat. Mach and velocity are NOT interchangeable.
clear; clc; close all;

%% Editable inputs
gamma = 1.4;                % cp/cv
R = 287;                    % Specific gas constant [J/(kg K)]
T0 = 300;                   % Stagnation temperature [K]
L = 1;                      % Nozzle length [m]
```

```
At = 0.01;                    % Geometric throat area [m^2]
inletRatio = 2.5;             % A_inlet / At (>1)
exitRatio = 3.0;              % A_exit / At (>1)
subsonicAstarRatio = 0.7;      % A*_subsonic / At (strictly between 0 and 1)
assert(gamma>1 && R>0 && T0>0 && L>0 && At>0);
assert(inletRatio>1 && exitRatio>1);
assert(subsonicAstarRatio>0 && subsonicAstarRatio<1);

%% Geometry: smooth area minimum at x = L/2
x = linspace(0,L,501);
xt = L/2;
A = zeros(size(x));
upstream = x <= xt;
A(upstream) = At*(1+(inletRatio-1)*((x(upstream)-xt)/xt).^2);
A(~upstream) = At*(1+(exitRatio-1)*((x(~upstream)-xt)/(L-xt)).^2);

%% Integrated form of Eq. 2.27 with isentropic perfect-gas relations
% A/A* = (1/M)*[2/(gamma+1)*(1+(gamma-1)*M^2/2)]^...
%         [(gamma+1)/(2*(gamma-1))]
% A* is the sonic reference area. It equals the physical throat area
% only for the choked solution. Back pressure must support the chosen
% shock-free transonic solution; geometry alone does not select it.
areaMach = @(M) (1./M).*(2/(gamma+1).*...
    (1+(gamma-1)/2.*M.^2)).^((gamma+1)/(2*(gamma-1)));
Mchoked = zeros(size(x));
Msubsonic = zeros(size(x));
for k = 1:numel(x)
    ratio = A(k)/At;
    if abs(ratio-1)<1e-12
        Mchoked(k) = 1;
    elseif x(k)<xt
        Mchoked(k) = fzero(@(M) areaMach(M)-ratio,[1e-8,1]);
    else
        upper = 2;
        while areaMach(upper)<ratio
            upper = 2*upper;
        end
        Mchoked(k) = fzero(@(M) areaMach(M)-ratio,[1,upper]);
    end
    ratioSub = A(k)/(subsonicAstarRatio*At);
    Msubsonic(k) = fzero(@(M) areaMach(M)-ratioSub,[1e-8,1]);
end

%% Static temperature, local sound speed, and actual velocity
Tchoked = T0./(1+(gamma-1)/2*Mchoked.^2);
Tsubsonic = T0./(1+(gamma-1)/2*Msubsonic.^2);
Vchoked = Mchoked.*sqrt(gamma*R*Tchoked);
Vsubsonic = Msubsonic.*sqrt(gamma*R*Tsubsonic);
assert(max(abs(areaMach(Mchoked)-A/At))<1e-7);
```

```
%% Plots: Mach on the y-axis; nozzle position on the x-axis
%% Clean, print-friendly plots
fig = figure('Color','w', ...
    'Units','inches', ...
    'Position',[1 1 7.5 9]);

if isprop(fig,'Theme')
    fig.Theme = 'Light';
end

tl = tiledlayout(fig,3,1, ...
    'TileSpacing','loose', ...
    'Padding','loose');

blue   = [0.00 0.35 0.70];
orange = [0.85 0.33 0.10];

% 1. Mach number
ax1 = nexttile(tl);
p1 = plot(x,Mchoked,'Color',blue,'LineWidth',2);
hold on;
p2 = plot(x,Msubsonic,'--','Color',orange,'LineWidth',2);

yline(1,':k','HandleVisibility','off');
xline(xt,':k','HandleVisibility','off');

ylabel('Mach number, M');
title('Mach number along the nozzle');
grid on;

lgd = legend([p1 p2], ...
    {'Choked, shock-free','Entirely subsonic'}, ...
    'Orientation','horizontal', ...
    'FontSize',11, ...
    'Box','off');
lgd.Layout.Tile = 'north';

% 2. Nozzle area
ax2 = nexttile(tl);
plot(x,A/At,'k-','LineWidth',2);
hold on;
xline(xt,':k','HandleVisibility','off');

ylabel('Area ratio, A/A_t');
title('Nozzle area distribution');
grid on;

% 3. Actual velocity
ax3 = nexttile(tl);
plot(x,Vchoked,'Color',blue,'LineWidth',2);
```

```
hold on;
plot(x,Vsubsonic,'--','Color',orange,'LineWidth',2);
xline(xt,':k','HandleVisibility','off');

ylabel('Velocity, V [m/s]');
xlabel('Nozzle position, x [m]');
title('Actual gas speed');
grid on;

% Consistent formatting
axs = [ax1 ax2 ax3];
set(axs, ...
    'FontSize',11, ...
    'LineWidth',1, ...
    'Box','on', ...
    'Color','w', ...
    'XColor','k', ...
    'YColor','k');

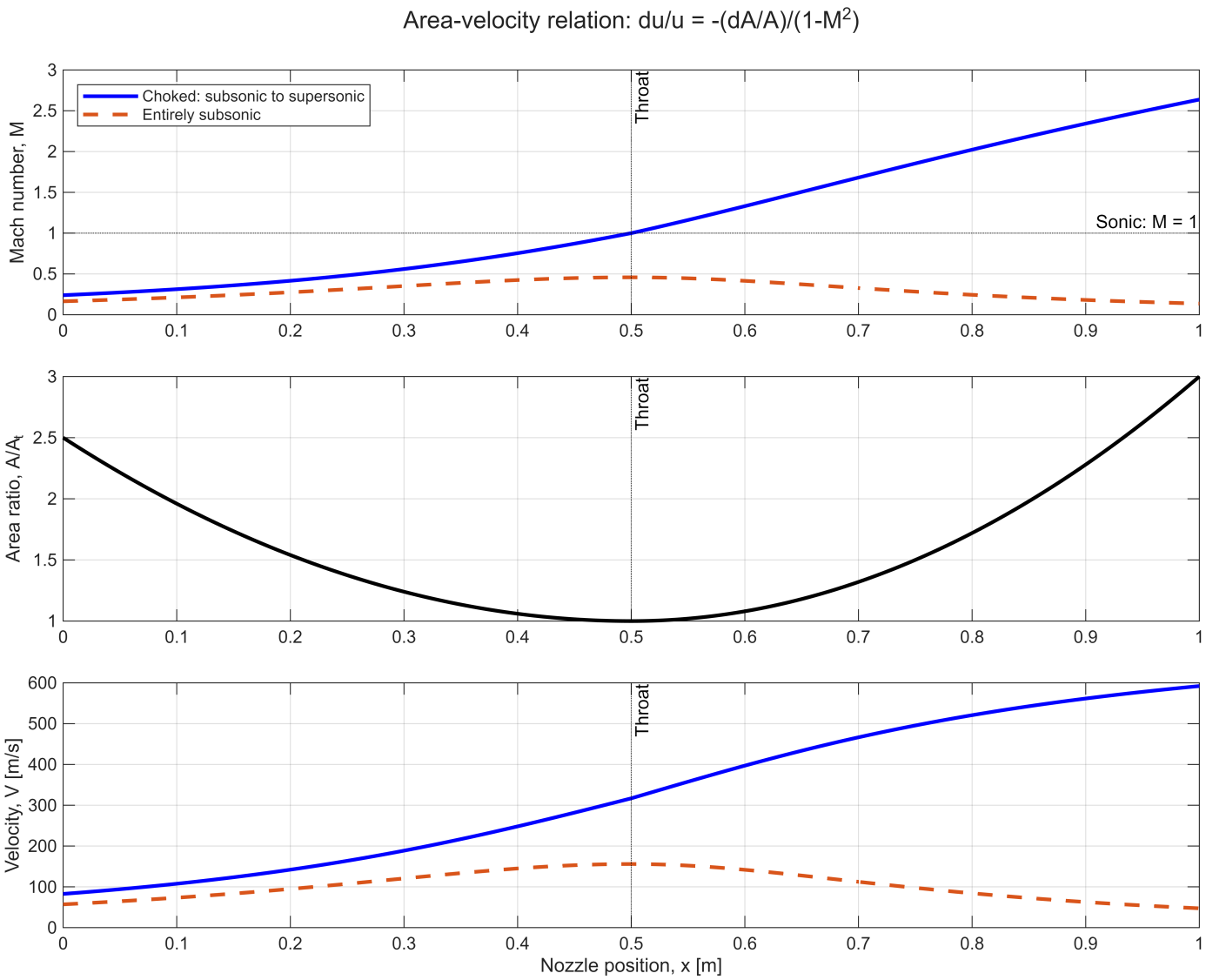
for ax = axs
    ax.Title.FontSize = 13;
    ax.Title.FontWeight = 'bold';
    ax.YLabel.FontSize = 12;
    ax.GridAlpha = 0.15;
end
ax3.XLabel.FontSize = 12;

linkaxes(axs,'x');
xlim(ax1,[0 L]);

% Only the bottom plot needs x-axis tick labels
ax1.XTickLabel = [];
ax2.XTickLabel = [];

drawnow;

% Export the entire layout, including the external legend
exportgraphics(tl,'Nozzle_Area_Mach.pdf', ...
    'ContentType','vector', ...
    'BackgroundColor','white');
```




```
Choked case: inlet M = 0.240, throat M = 1, exit M = 2.637
Choked case: inlet V = 82.7, exit V = 592.2 m/s
```

The first chart, 'Mach number along a converging-diverging nozzle', shows two curves - a choked: subsonic to supersonic Mach number, reaching M=1 at the throat, then continuing to increase. The second curve, is entirely subsonic, past the throat (with increasing area) the Mach number shows a gradual decrease as expected in a "subsonic world."

Problem 2-B. Supersonic Flow

Problem Statement

List every indication of supersonic flow described in the movie.

Statement

Appearance of shock waves in the channel means that the flow is locally supersonic.

In a one-dimensional approximation, the intersection of theoretical pressure curves for the channel marks the conditions for which air accelerating from rest should reach sonic speed.

When the researcher introduced artificial roughness, or tape to simulate roughness, Mach waves originate at the edges of the tape, which are evidence of supersonic flow in the diverging part of the channel.

Pressure distribution is also an indicator of supersonic flow, in the diverging channel, at first the pressure is decreasing, and hence the velocity increases. However, after the shock wave, the pressure increases, and the velocity decreases, showing subsonic flow. The supersonic region ends at the shock wave [1].

Problem 3. Stagnation Enthalpy

Problem Statement

A meteor enters the earth's atmosphere at 35,000 mph. What is the stagnation enthalpy on the meteor?

Analysis

Assumption: Analysis is based off the meteor atmospheric entry at the Karman line (altitude above MSL: 100 km, 62 miles) with the following temperature conditions:

- $T_{Karman} = 195\,^{\circ}K (-78\,^{\circ}C, -108\,^{\circ}F)$

With hypersonic flight, atmospheric entry, we reference absolute zero, $T_{ref} = 0\,^{\circ}K$. At $0\,^{\circ}K$, molecular motion stops, so enthalpy is defined as exactly $0\,kJ/kg$.

$T = T_{Karman} - T_{ref} = 195\,^{\circ}K - 0\,^{\circ}K$

Enthalpy at the Karman line:

$h = c_p T = 1.004\,kJ/(kgK) \times 195\,^{\circ}K = 195.78\,kJ/kg$

Stagnation enthalpy on the meteor:

$h_0 = h + 1/2u^2 = 195,780\,J/kg + 1/2(15,646 m/s)^2 = 122,594,438\,J/kg = \mathbf{122.6}\ \ \mathbf{MJ/kg}$

- Note: 99.8% of the stagnation enthalpy on the meteor is from the meteor's kinetic energy, and only 0.2% is attributed to the static enthalpy.

Physical Interpretation

Stagnation enthalpy is used by engineers as an energy reference, or an "energy budget" against which to calculate changing local conditions. In other words, any downstream local enthalpy conditions will be less than the stagnation enthalpy [2].

References

[1] YouTube '8. Channel Flow of a Compressible Fluid'.

[2] Cengel, Y. (2018). *Fluid Mechanics: Fundamentals and Applications* (4th ed.). New York: McGraw-Hill.

[3] Liepmann, H. W., & Roshko, A. (2001). *Elements of gasdynamics*. Dover Publications. (Original work published 1957)